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Bihar Engineering University, Patna

B.Tech 4th Semester Exam., 2022

Code: 101403 ***Time: 03 Hours***

Semester: IV ***Subject: Engineering Geology (New Course)*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (any seven): **[2 x 7 = 14]**

(a) Which is the longest dam in India?
(i) Hirakud dam
(ii) Bhakra dam
(iii) Krishna Raja Sagar dam
(iv) Nagarjuna Sagar dam

(b) The measure of the volume of water that can be drawn from a given reservoir in a certain interval of time is called
(i) surface yield
(ii) drawable yield
(iii) reservoir yield
(iv) pool yield

(c) Rocks formed from molten magma are known as
(i) sedimentary rocks
(ii) igneous rocks
(iii) metamorphic rocks
(iv) None of the above

(d) Limestone is not a
(i) sedimentary rock
(ii) stratified rock
(iii) aqueous rock
(iv) metamorphic rock

(e) The softest and hardest rocks (in that order) are
(i) marble and diamond
(ii) gypsum and orthoclase
(iii) talc and diamond
(iv) quartz and topaz

(f) The mineral which is almost black but gives brown streak is
(i) magnetite
(ii) garnet
(iii) hornblende
(iv) chromite

(g) Fractures along which there has been no relative displacement are called
(i) Faults
(ii) Joints
(iii) Folds
(iv) Intrusions

(h) The type of fault where the faulted blocks have been moved against each other in horizontal direction is
(i) reverse fault
(ii) hinge fault
(iii) strike-slip fault
(iv) vertical fault

(i) What is the other name for upright fold?
(i) Symmetrical fold
(ii) Asymmetrical fold
(iii) Overturned fold
(iv) Isoclinal fold

(j) Which of the following theories are related to the origin of the earth?
(i) The tidal hypothesis
(ii) The nebular hypothesis
(iii) The planetesimal hypothesis
(iv) All of the above

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Q.2 Write the answers in short: **[7 x 2 = 14]**

(a) What is the scope of geology in civil engineering? Describe the internal structure of the earth.

(b) Write notes on prevention, control and correction of landslides.

Q.3 Write the answers in short: **[7 x 2 = 14]**

(a) What is an aggregate? Describe the engineering properties of rocks as an aggregate for road construction.

(b) Write down the seismic considerations for designing an engineered construction.

Q.4 Write the answers in short: **[7 x 2 = 14]**

(a) Describe the various sources of ground water. Write a note on the factors controlling the movement of ground water. Discuss ground water hazards in engineering projects.

(b) With a neat sketch illustrate the different layers of a weathered zone under normal circumstances.

Q.5 Write the answers in short: **[7 x 2 = 14]**

(a) What are the various forces acting on a gravity dam and an arch dam? What are the requisite qualities of the dam site to ensure stability of dam in view of these forces?

(b) Give an account of important factors to be considered for evolving a seismic design in a seismic region.

Q.6 Write the answers in short: **[7 x 2 = 14]**

(a) Write a brief note on "joints in rocks, their causes and effects on the engineering quality of rocks".

(b) Explain the different types of the cut and cover method.

Q.7 Write the answers in short: **[7 x 2 = 14]**

(a) Explain the geological conditions that influence a tunnelling project.

(b) Occurrence of bed rock at shallow depth and narrow river valley are also important factors in selection of a dam site. Why?

Q.8 Write the answers in short: **[7 x 2 = 14]**

(a) In addition to the destruction caused directly by seismic vibrations, list three types of destructions associated with earthquakes.

(b) What do you understand by triggering mechanism in landslides?

Q.9 Write short notes on any four of the following: **[3½ x 4 = 14]**

(a) Solifluction deposits

(b) Inliers and outliers

(c) Strength of igneous rock structures

(d) Structural geology of rocks

(e) Waterfall and Gorges

(f) Mudstone and shale

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